



02 — Position, Velocity, Acceleration as Vectors: The Story of Motion

Kleppner: Ch 1.5~1.7 | **Time:** 70 min

Core Question: How do we extend the 1D idea "change-in-a-blink = velocity" into three dimensions?

Scene: A Drone Tracing a Curve Through the Sky

A drone flies in a figure-eight pattern. Its position changes every instant, and the direction of its velocity changes too.

How do we capture this motion in a single equation?

✗ What If You Try With Algebra Alone?

Why Must We Wrestle With This?

Nature changes ceaselessly. Velocity changes at every instant, and the change itself changes too. But the mathematics we know (algebra) can only handle **fixed values**.

"Average velocity" crushes an entire interval into a single number. But real physics asks: **"Right at this instant, how fast is it moving?"** To answer this question, you need a way of thinking that captures change as it changes. That is the change-in-a-blink.

Attempt: Split the Interval Finer and Finer

Suppose a drone's position is given by $\vec{r}(t) = (t^2, t^3)$. We want the **instantaneous velocity** at $t = 2$ seconds.

Pick clean fractions of a second: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$.

Interval = 1: from $t = 2$ to $t = 3$

$$\vec{r}(3) = (3^2, 3^3) = (9, 27)$$

$$\vec{v}_{\text{rough}} = \frac{(9, 27) - (4, 8)}{1} = \left(\frac{5}{1}, \frac{19}{1} \right) = (5, 19)$$

Interval = $\frac{1}{2}$: from $t = 2$ to $t = 2 + \frac{1}{2} = \frac{5}{2}$

$$\vec{r}\left(\frac{5}{2}\right) = \left(\frac{25}{4}, \frac{125}{8} \right)$$

$$\vec{v} = \frac{\left(\frac{25}{4}, \frac{125}{8} \right) - (4, 8)}{1/2} = \frac{\left(\frac{9}{4}, \frac{61}{8} \right)}{1/2} = \left(\frac{9}{2}, \frac{61}{4} \right) = (4.5, 15.25)$$

Interval = $\frac{1}{4}$: from $t = 2$ to $t = 2 + \frac{1}{4} = \frac{9}{4}$

$$\vec{r}\left(\frac{9}{4}\right) = \left(\frac{81}{16}, \frac{729}{64} \right)$$

$$\vec{v} = \frac{\left(\frac{81}{16}, \frac{729}{64} \right) - \left(\frac{64}{16}, \frac{512}{64} \right)}{1/4} = \frac{\left(\frac{17}{16}, \frac{217}{64} \right)}{1/4} = \left(\frac{17}{4}, \frac{217}{16} \right) = (4.25, 13.5625)$$

Interval = $\frac{1}{8}$: from $t = 2$ to $t = 2 + \frac{1}{8} = \frac{17}{8}$

After similar algebra (try it yourself!): $\vec{v} \approx (4.125, 12.7656)$

Interval = $\frac{1}{16}$: from $t = 2$ to $t = 2 + \frac{1}{16} = \frac{33}{16}$

After similar algebra: $\vec{v} \approx (4.0625, 12.3789)$

Interval	Approximate $\vec{v}$	Distance from (4, 12)
1	(5, 19)	(1, 7)
$1/2$	(4.5, 15.25)	(0.5, 3.25)
$1/4$	(4.25, 13.56)	(0.25, 1.56)
$1/8$	(4.125, 12.77)	(0.125, 0.77)
$1/16$	(4.0625, 12.38)	(0.0625, 0.38)

The pattern is clear: each time we **halve** the interval, the error is roughly halved too. But — critically — **the error never reaches zero**. With any finite interval, you can only get closer. To hit $(4, 12)$ exactly, you need a leap: **erase the interval entirely**. That leap is the limit.

✓ Resolved Through Limits

From Approximation to Exact: The Leap

The table shows a sequence approaching $(4, 12)$. Now ask: **"If the interval becomes zero, what value does the sequence reach?"**

That question is the limit. Writing it formally:

$$\vec{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} = \frac{d\vec{r}}{dt}$$

Now, instead of computing endlessly, we use the **rule for change-in-a-blink** that we already know. For $\vec{r}(t) = (t^2, t^3)$:

The x -component: $x(t) = t^2$. Its rate of change at any t is $2t$. Why? We can verify algebraically:

$$\frac{(t + \Delta t)^2 - t^2}{\Delta t} = \frac{t^2 + 2t\Delta t + (\Delta t)^2 - t^2}{\Delta t} = \frac{2t\Delta t + (\Delta t)^2}{\Delta t} = 2t + \Delta t \xrightarrow{\Delta t \rightarrow 0} 2t$$

The y -component: $y(t) = t^3$. Expanding:

$$\frac{(t + \Delta t)^3 - t^3}{\Delta t} = \frac{3t^2\Delta t + 3t(\Delta t)^2 + (\Delta t)^3}{\Delta t} = 3t^2 + 3t\Delta t + (\Delta t)^2 \xrightarrow{\Delta t \rightarrow 0} 3t^2$$

So the rule is: $\frac{d}{dt}(t^2, t^3) = (2t, 3t^2)$.

At $t = 2$: $\vec{v} = (4, 12)$ — **clean, exact, arrived at in one step**.

What This Means

Nature speaks in "change-in-a-blink." The rules we learned — $x'(t) = 2t$, $\frac{d}{dt}t^3 = 3t^2$ — are not arbitrary formulas. They are the result of asking: "What happens when the interval truly vanishes?" Each rule is the fingerprint of that limit.

Each Axis Is Independent

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

- x -direction: only $x(t)$ matters. y -direction: only $y(t)$ matters.
 - **In a horizontal throw: horizontal is constant velocity, vertical is free-fall — completely independent.**
 - Acceleration: $\vec{a} = \frac{d\vec{v}}{dt} = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2} \right)$
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Examples — Reading Motion Through Change-in-a-Blink

Example 1: Uniform Straight-Line Motion

$$\vec{r}(t) = (3t, 4t) \rightarrow \vec{v} = (3, 4), \vec{a} = (0, 0)$$

"Every second: 3 m in x , 4 m in y , unchanged. No acceleration whatsoever."

Example 2: Constant-Acceleration Motion

$$\vec{r}(t) = (2t, 5t - 5t^2) \rightarrow \vec{v} = (2, 5 - 10t), \vec{a} = (0, -10)$$

"Horizontal: constant 2 m/s. Vertical: accelerating downward at 10 m/s². Projectile motion!" (Using $g = 10$ for simplicity.)

Example 3: Circular Motion

$$\vec{r}(t) = (R \cos \omega t, R \sin \omega t) \rightarrow \vec{v} = (-R\omega \sin \omega t, R\omega \cos \omega t) \rightarrow \vec{a} = (-R\omega^2 \cos \omega t, -R\omega^2 \sin \omega t) = -\omega^2 \vec{r}$$

"The velocity vector leads the position vector by 90°. Acceleration always points toward the center."

Example 4: Helical Motion

$$\vec{r}(t) = (\cos t, \sin t, t) \rightarrow \vec{v} = (-\sin t, \cos t, 1) \rightarrow \vec{a} = (-\cos t, -\sin t, 0)$$

"Circular motion in the xy -plane, steady climb in z . Acceleration is purely centripetal."

Example 5: Detecting Rest via Change-in-a-Blink

$$\vec{r}(t) = (t^3 - 3t, t^2 - 2t) \rightarrow \vec{v} = (3t^2 - 3, 2t - 2)$$

At $t = 1$: $\vec{v} = (0, 0)$ — momentarily at rest!

But $\vec{a} = (6t, 2) = (6, 2)$ — acceleration is still present. It will start moving again immediately.

Example 6: Reconstructing a Trajectory From Initial Conditions

$$\vec{a} = (0, -10), \vec{v}_0 = (20, 30), \vec{r}_0 = (0, 0)$$

$$\rightarrow \vec{v}(t) = (20, 30 - 10t)$$

$$\rightarrow \vec{r}(t) = (20t, 30t - 5t^2)$$

"Gather the change-in-a-blink of velocity and position is restored. The path is open in the reverse direction too."

Meaning: "What We Just Did"

When the 1D concept "change-in-a-blink = velocity, change-in-a-blink of velocity = acceleration" extends to 3D, the fact that **each axis is independent** makes everything simple.

With algebra alone, you can only forever approximate the instantaneous velocity of $\vec{r}(t) = (t^2, t^3)$. The moment you introduce the limit $\Delta t \rightarrow 0$, the exact value $(4, 12)$ emerges in one stroke.

"Capturing the change-in-a-blink — this is the first letter of the language that describes motion."



Basic Drills (*solutions in solutions/02-solutions.md*)

1. For $\vec{r}(t) = (4t, 3t^2)$, find velocity and acceleration at $t = 2$.
2. For $\vec{r}(t) = (5 \cos 2t, 5 \sin 2t)$, is the magnitude of velocity constant? What is the direction of acceleration?
3. Given $\vec{v}(t) = (6t, 8)$ and $\vec{r}(0) = (0, 0)$, find $\vec{r}(3)$.
4. Given $\vec{a} = (0, -10)$, $\vec{v}_0 = (10, 20)$, $\vec{r}_0 = (0, 30)$, find $\vec{r}(t)$.
5. For $\vec{r}(t) = (t^2, t^3 - t)$, find the time t when the object is momentarily at rest.



Advanced Drills (*solutions in solutions/02-solutions.md*)

1. For $\vec{r}(t) = (e^t \cos t, e^t \sin t)$, find velocity and acceleration. What shape is this motion?
2. For $\vec{r}(t) = (t \cos t, t \sin t)$, find the speed $|\vec{v}|$ at $t = 2\pi$.
3. Projectile motion: $\vec{r}(t) = (v_0 \cos \theta \cdot t, v_0 \sin \theta \cdot t - \frac{1}{2}gt^2)$. Find $\vec{v}(t)$ and $\vec{a}(t)$, and prove that $|\vec{v}|$ is minimized at the peak.
4. Given $\vec{a}(t) = (\cos t, -\sin t, 1)$, $\vec{v}(0) = (1, 0, 0)$, $\vec{r}(0) = (0, 1, 0)$, find $\vec{r}(t)$. (Warning: two rounds of the reverse of change-in-a-blink.)
5. **Trap:** For $\vec{r}(t) = (t^2, t^4)$, at what t does $|\vec{v}|$ become zero? And at that instant, is the acceleration zero?



Intuition Drills (*solutions in solutions/02-solutions.md*)

1. A particle's acceleration vector is **always perpendicular** to its velocity vector. What can you conclude about the speed?
2. A particle is at its **maximum height** in projectile motion. At this instant, is its acceleration zero? If not, what is it?
3. Two identical balls are launched from a cliff: one is dropped straight down, the other is fired horizontally. Which hits the ground first?
4. A particle moves with $\vec{v}(t) = (3, 4)$ and starts at $\vec{r}(0) = (0, 0)$. What is the shape of its path?

5. A particle starts from rest with $\vec{a} = (2, 0)$ for $t = 0$ to $t = 3$, then $\vec{a} = (0, -2)$ for $t = 3$ to $t = 6$. Sketch the path. What shape is it?
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Reading Physical Symbols

Symbol	How to Read It	Physical Meaning
$d\vec{r}/dt$	"dee-r dee-t"	Change-in-a-blink. How much $\vec{r}$ changes in an infinitely small interval of time. The core idea of all kinematics.
$\vec{v}(t)$	"v of t"	Velocity at time t . A function : its value changes as t changes.
$\vec{a}(t)$	"a of t"	Acceleration at time t . The change-in-a-blink of the change-in-a-blink of position.
$\lim_{\Delta t \rightarrow 0}$	"limit as delta-t goes to zero"	The operation that erases the interval . What remains is the exact instantaneous value.
$\vec{r}_0, \vec{v}_0$	"r-nought, v-nought"	Initial conditions — where the object starts and how fast it's moving at $t = 0$. Without these, the future is not uniquely determined.
$x(t), y(t), z(t)$	"x of t, y of t, z of t"	The three independent stories of motion. Each axis evolves on its own.

Key habit: When you see $d\vec{r}/dt$, don't just compute — visualize the arrow $\vec{r}$ getting a tiny nudge at its tip. That nudge *is* the velocity.

Glossary